Neurodevelopmental Disorder With Poor Growth, Large Ears, and Dysmorphic Facies (NEDGDF / ZNF668-related ZMAND)

MONDO:0859350 · OMIM:620194 · Gene: ZNF668 (HGNC:25821)

Comprehensive disease-knowledge-base report. Evidence base is small (one defining publication describing 3 affected individuals from 2 consanguineous families, plus in-vitro cancer-biology literature on ZNF668 function). Evidence-source types are flagged throughout as **[human clinical]**, **[in vitro]**, or **[computational]**.

Summary (Answer to the Research Question)

Neurodevelopmental Disorder With Poor Growth, Large Ears, and Dysmorphic Facies (NEDGDF; proposed clinical name **ZNF668-related ZMAND** — *ZNF668-related brain malformation, microcephaly, abnormal growth, neurodevelopmental delay, and dysmorphism*) is an ultra-rare **autosomal-recessive Mendelian disorder of the DNA-damage-repair (DDR) machinery** caused by **biallelic truncating (loss-of-function) variants in *ZNF668*** (zinc finger protein 668), a nucleolar C2H2 zinc-finger protein on chromosome 16p11.2. Loss of ZNF668 impairs Tip60(KAT5)-dependent histone H2AX acetylation/chromatin relaxation and homologous-recombination repair, and destabilizes p53 via MDM2, producing genome instability. Clinically it presents in infancy with prenatal-onset growth restriction (small for gestational age, failure to thrive), microcephaly, severe global developmental delay/intellectual disability, hypotonia, brain malformation/atrophy, and a recognizable facial gestalt (large ears/macrotia, prominent nose, deep-set eyes). It was delineated by Alsaif et al., 2021 (P 34313816).

There is no specific therapy; management is supportive.

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1. Disease Information

Overview. A recognizable autosomal-recessive multisystem neurodevelopmental/dysmorphic syndrome in the growing class of **Mendelian DNA-damage-repair disorders**. First formally delineated as a distinct entity in 2021 **[human clinical]**: *"We present new evidence of a distinct disorder caused by biallelic truncating variants in ZNF668 comprising microcephaly, growth deficiency, severe global developmental delay, brain malformation, and distinct facial dysmorphism."* (Alsaif et al.,).

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Key identifiers. - **MONDO:** MONDO:0859350 - **OMIM (phenotype):** 620194 - **OMIM (gene):** 617103 (*ZNF668*) - **MedGen:** C5774288 / CUI 1824061; **UMLS:** C5774288 - **Orphanet:** No specific ORPHAcode identified at time of writing (not available / likely pending). - **ICD-10:** No specific code; best fit Q87.8 (other specified congenital malformation syndromes) / within Q87 + associated codes for microcephaly (Q02) and intellectual disability (F7x). **ICD-11:** best fit LD2F.1Y / LD90.Y (other specified syndromes with CNS/developmental anomalies) — no bespoke code. - **MeSH:** No specific descriptor; indexed under Neurodevelopmental Disorders (D065886), Intellectual Disability (D008607), Microcephaly (D008831), DNA Repair-Deficiency Disorders (D049914).

Synonyms / alternative names. - Neurodevelopmental disorder with poor growth, large ears, and dysmorphic facies (NEDGDF) - ZNF668-related ZMAND (ZNF668-related brain malformation, microcephaly, abnormal growth, neurodevelopmental delay, and dysmorphism) - ZNF668 deficiency / ZNF668-related DNA-damage-repair disorder

Information source. Disease-level aggregated resources (OMIM, MONDO, HPO) **derived from individual-patient case reports** in a single primary publication (; 3 patients). Not EHR-derived at population scale.

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2. Etiology

Primary cause — genetic. Biallelic (homozygous or compound-heterozygous) **truncating loss-of-function variants in *ZNF668* [human clinical]** (P 34313816).

This is a monogenic, fully genetic etiology; no environmental or infectious cause.

Genetic risk factors. - **Causal variants:** biallelic truncating (nonsense/frameshift) *ZNF668* variants. Both reported families were **consanguineous** → homozygous variants by descent. - **Susceptibility loci / modifier genes:** none established (n too small).

Environmental risk factors. The only recognized "risk factor" is **parental consanguinity / shared ancestry** increasing homozygosity for rare recessive alleles. No toxin, radiation, lifestyle, or occupational exposure is implicated as a cause. (Note: because the protein participates in repair of exogenous DNA damage, DNA-damaging exposures such as ionizing/UV radiation are mechanistically relevant stressors in vitro, but are **not** disease causes.)

Protective factors. None identified. Being a **heterozygous carrier** is protective relative to biallelic loss (carriers are unaffected); i.e., a single functional allele is sufficient (recessive).

Gene–environment interactions. Not characterized. Plausible but undemonstrated interaction between residual DDR capacity and genotoxic exposure (**inferred**, not shown in patients).

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3. Phenotypes

Phenotype frequencies below are from HPO annotation of OMIM:620194 (source P 34313816 ; denominators reflect the 3 reported individuals; **[human clinical]**). Onset is **infantile/congenital** (HP:0003593 Infantile onset, 3/3). Course is chronic and non-remitting; severity of neurodevelopmental impairment is **severe**. Quality-of-life impact is **profound** given severe developmental delay/intellectual disability, hypotonia, feeding difficulties, and seizures, implying lifelong dependence for daily activities (no formal EQ-5D/SF-36 data exist for this ultra-rare disease).

Invariant (3/3 ≈ 100%) — physical/growth/neuro signs: - Macrotia / large ears — **HP:0000400** - Microcephaly — **HP:0000252** - Prominent nose — **HP:0000448** - Small for gestational age (prenatal growth restriction) — **HP:0001518** - Failure to thrive — **HP:0001508** - Global developmental delay — **HP:0001263** - Hypotonia — **HP:0001252** - Long hallux — **HP:0001847** - Infantile onset — **HP:0003593**

Frequent (2/3 ≈ 67%; intellectual disability 2/2): - Intellectual disability — **HP:0001249** - Seizure — **HP:0001250** - Deeply set eyes — **HP:0000490** - High palate — **HP:0000218** - Preauricular skin tag — **HP:0000384** - Hyperplasia of the maxilla — **HP:0430028** - Hallux valgus — **HP:0001822**

Variable (1/3 ≈ 33%; osteopenia 1/2): - Secundum atrial septal defect — **HP:0001684** (cardiac) - Chorioretinal coloboma — **HP:0000567** (ophthalmologic) - Micrognathia — **HP:0000347** - Brain atrophy — **HP:0012444** - Epileptic encephalopathy — **HP:0200134** - Feeding difficulties — **HP:0011968** - Recurrent lower respiratory tract infections — **HP:0002783** - Diabetes mellitus — **HP:0000819** (laboratory/metabolic abnormality) - Lower-limb spasticity — **HP:0002061** - Pectus carinatum — **HP:0000768** - Broad thumb — **HP:0011304** - Clinodactyly — **HP:0030084** - Osteopenia — **HP:0000938** (laboratory/imaging abnormality)

Phenotype-type classification. Physical/dysmorphic manifestations (ears, nose, eyes, palate, jaw, digits, chest), clinical neurological signs (microcephaly, hypotonia, spasticity, seizures/encephalopathy, brain atrophy), developmental/behavioral (developmental delay, intellectual disability), and laboratory abnormalities (diabetes mellitus, osteopenia). Brain malformation and DDR defect are additional features noted in the abstract ( ).

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4. Genetic / Molecular Information

Causal gene. ***ZNF668*** — zinc finger protein 668. HGNC:25821; NCBI Gene 79759; OMIM gene 617103; Ensembl ENSG00000167394; canonical transcript ENST00000300849; UniProt Q96K58. Location **16p11.2** (GRCh38 chr16:31,059,772–31,074,240, minus strand). Protein: C2H2-type Krüppel-like zinc-finger (KRAB-less) transcription factor; predicted sequence-specific DNA-binding / RNA Pol II regulator; **nucleolar** localization ( ).

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[computational/in vitro]

Pathogenic variants. Clinical RefSeq transcript **NM_024706.5** (protein NP_078982). - **Type/class:** biallelic **truncating** variants (nonsense and/or frameshift) → premature termination / NMD → **loss of function** (34313816). No recurrent pathogenic missense established for the disorder. - **Representative ClinVar pathogenic/likely-pathogenic truncating alleles** (NM_024706.5): **c.1111C>T (p.Arg371Ter)** — Pathogenic (nonsense); **c.955C>T (p.Gln319Ter)** — Pathogenic (nonsense); **c.1197del (p.Lys399fs)** — Likely pathogenic (frameshift); c.1517dup (p.Gly506_Glu507insTer) — currently VUS. The **majority of ~50 ZNF668 ClinVar records are missense/synonymous variants classified VUS or (likely) benign**, reinforcing that pathogenicity requires truncation/null rather than missense. **[computational/ClinVar]** - **Classification (ACMG/AMP):** disease-causing variants are **pathogenic/likely pathogenic** truncating alleles segregating with disease in consanguineous families; a null variant in this LoF-intolerant gene meets the strong pathogenic criterion (PVS1). - **IMPORTANT distinction — 16p11.2 CNVs:** *ZNF668* lies within the recurrent **16p11.2** region. Numerous ClinVar "Pathogenic" entries at 16p11.2 are **large contiguous-gene deletions/duplications** (e.g., GRCh37 chr16:30,851,860–31,156,762 x1) that encompass *ZNF668* among many genes; these represent the **dominant 16p11.2 microdeletion/microduplication syndrome**, a *distinct* disorder that must **not** be conflated with the autosomal-recessive single-gene ZNF668-ZMAND entity when interpreting chromosomal microarray. - **Zygosity / origin: germline, homozygous** (both families consanguineous). No somatic/mosaic disease origin reported for the Mendelian disorder. - **Functional consequence: loss of function** (haploinsufficiency is tolerated — carriers unaffected — so biallelic complete LoF is required).

Population allele frequency / constraint (gnomAD v2.1.1, computational). *ZNF668* is strongly **loss-of-function intolerant at the heterozygous level**: **pLI = 0.999**, **LOEUF = 0.42**, observed/expected LoF = 10/40.3 (**oe = 0.25**), **lof_z = 4.05**; missense **mis_z = 2.21**. Disease-causing truncating alleles are ultra-rare/absent in population databases. (Note the apparent paradox: strong heterozygous LoF constraint yet recessive inheritance — consistent with dosage sensitivity but sufficiency of one allele to prevent the syndrome.)

Somatic relevance (cancer). *ZNF668* is a **somatically mutated breast-cancer tumor-suppressor gene** (genome-wide sequencing identified frequent mutations; 21852383) and shows **reduced expression associated with bladder-cancer invasiveness**

(P 37240013)

[in vitro / tumor genomics] — relevant to protein function, distinct from the germline recessive syndrome.

Modifier genes / epigenetics / chromosomal abnormalities. No disease modifiers, constitutional methylation signature, or recurrent chromosomal abnormality (CNV/translocation) defined for NEDGDF. (ZNF668 itself regulates chromatin acetylation — see Mechanism.) Not applicable / not available.

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5. Environmental Information

- **Environmental factors:** None causal. The disorder is monogenic. DNA-damaging agents (ionizing radiation, UV) are mechanistically relevant **stressors** that expose the repair defect in cells (P 23777805) but are not disease causes. **[in vitro]**
- **Lifestyle factors:** Not applicable (congenital genetic disease).
- **Infectious agents:** None. (Recurrent lower respiratory tract infections, HP:0002783, occur as a *secondary complication* of the neurodevelopmental phenotype, not as a cause.)

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6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation)

1. **Biallelic truncating variant in *ZNF668*** leads to loss of full-length nucleolar ZNF668 protein (NMD/truncation). **[human clinical,** P 34313816 **]**
2. Loss of ZNF668 **results in** two converging molecular deficits:
3. **Branch A (chromatin/repair):** loss of ZNF668–**Tip60 (KAT5)** cooperation **leads to** reduced ionizing-radiation-induced **histone H2AX acetylation** and **impaired chromatin**

relaxation at double-strand breaks. **[in vitro,**

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P 23777805

4. **Branch B (p53 axis):** loss of ZNF668 regulation of **MDM2 leads to** failure to stabilize **p53** after DNA damage (ZNF668 normally restrains MDM2-mediated p53 ubiquitination/degradation). **[in vitro,**

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5. Branch A **results in** decreased recruitment of homologous-recombination (HR) repair factors (and reduced RPA phosphorylation/recruitment after UV) → **defective HR / nucleotide-excision-type repair** and reduced cell survival after DNA damage. **[in vitro,**

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6. Branches A+B together **result in genome instability / accumulated unrepaired DNA damage** and a dysregulated DNA-damage checkpoint/apoptosis balance (demonstrated as a **DNA-damage-repair defect in patient fibroblasts**). **[human clinical,**

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7. Genome instability in **highly proliferative progenitors** (neural progenitors, somatic growth tissues) **leads to** impaired proliferation/increased cell death during development (**inferred**, by analogy to other microcephalic DDR/primordial-growth disorders,

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P 19473885

8. This **results in** the clinical phenotype: **microcephaly** + **brain malformation/atrophy** (neural progenitor depletion), **prenatal-onset growth restriction / SGA / failure to thrive** (reduced somatic proliferation), **severe developmental delay/intellectual disability, hypotonia, seizures**, and the **dysmorphic facial/skeletal gestalt** (large ears, prominent nose, deep-set eyes, palate/jaw and digit anomalies). **[human clinical,**

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Upstream vs downstream. Upstream = *ZNF668* LoF → chromatin-acetylation/HR-repair defect + p53 destabilization (molecular). Downstream = genome instability → progenitor attrition → microcephaly/growth failure/neurodevelopmental impairment (organ/clinical).

Checklist detail. - **Molecular pathways:** DNA-damage response; homologous-recombination DSB repair; p53–MDM2 tumor-suppressor axis; chromatin (histone H2AX/H2A) acetylation by Tip60/KAT5. (Reactome: DNA Double-Strand Break Repair; HDR through HR; Regulation of

TP53 Activity.) - **Cellular processes:** DNA repair, cell-cycle checkpoint control, apoptosis/survival after genotoxic stress, chromatin remodeling, (neuro)progenitor proliferation. - **Protein dysfunction:** complete **loss of function** of a nucleolar C2H2 zinc-finger transcriptional/chromatin regulator (truncation removes zinc-finger/functional domains). - **Metabolic / immune:** no primary metabolic pathway defect (diabetes mellitus is a variable late feature; mechanism unclear). No primary immunodeficiency (recurrent LRTIs are secondary to neurologic status). - **Molecular profiling (disease-specific):** no patient transcriptomic/proteomic/metabolomic/epigenomic datasets published. Functional assay = DDR defect in patient fibroblasts (P 34313816); RNAi/knockdown functional genomics in U2OS and mammary epithelial cells (PMIDs 23777805, 21852383).

Suggested GO / CL terms. GO:0006281 DNA repair; GO:0000724 double-strand break repair via HR; GO:0006325 chromatin organization; GO:0043968 histone H2A acetylation (and GO:0016573 histone acetylation); GO:0000077 DNA damage checkpoint signaling; GO:0006974 DNA-damage response; GO:0030578/GO:2000774 positive regulation of p53. Cell types (CL): CL:0000031 neuroblast (neural progenitor), CL:0000047 neuronal stem cell, CL:0002319 neural cell, CL:0000057 fibroblast.

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7. Anatomical Structures Affected

Organ / body-system level. - **Central nervous system (primary):** brain — microcephaly, brain malformation, brain atrophy; nervous system disorder with hypotonia, spasticity, seizures/epileptic encephalopathy. UBERON:0000955 brain; UBERON:0001890 forebrain; UBERON:0001017 central nervous system. - **Growth / musculoskeletal (primary):** generalized growth failure; skeletal features — maxilla (UBERON:0002397), palate (UBERON:0001716), mandible/jaw (micrognathia; UBERON:0001684), thorax (pectus carinatum; UBERON:0002228 rib cage), hands/feet (broad thumb, clinodactyly, long hallux, hallux valgus; UBERON:0002389 manual digit / UBERON:0001466 pedal digit); osteopenia (reduced bone mineral density). - **Craniofacial / ear / eye (dysmorphism):** external ear/pinna (macrotia, preauricular tag; UBERON:0001690 ear, UBERON:0001757 pinna), nose (UBERON:0000004), orbit/eye (deeply set eyes; chorioretinal coloboma — UBERON:0000479 retina/choroid). - **Cardiovascular (secondary/variable):** heart — secundum atrial septal defect (UBERON:0000948 heart; interatrial septum). - **Respiratory (secondary):** lower respiratory tract — recurrent infections

(UBERON:0001558 lower respiratory tract). - **Digestive (secondary):** feeding difficulties/failure to thrive (UBERON:0001007 digestive system). - **Endocrine/metabolic (variable):** diabetes mellitus (pancreas/glucose homeostasis; UBERON:0001264 pancreas).

Tissue / cell level. Nervous tissue (neurons, neural progenitors) and rapidly dividing somatic/connective tissues are most affected by the proliferative/genome-instability defect; fibroblasts show the measurable repair defect **[human clinical]**. CL terms: CL:0000031 neuroblast, CL:0000047 neuronal stem cell, CL:0000057 fibroblast, CL:0000679 glutamatergic neuron (general neuronal involvement).

Subcellular level. Nucleus (GO:0005634) and specifically the **nucleolus** (GO:0005730) — ZNF668 is a nucleolar protein (P 21852383); sites of DNA double-strand breaks / chromatin (GO:0000785 chromatin; GO:0035861 site of DNA damage).

Localization / lateralization. Craniofacial and CNS involvement is **bilateral/symmetric**; chorioretinal coloboma and limb anomalies may be variable/asymmetric; ASD is a midline cardiac lesion.

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8. Temporal Development

- **Onset: congenital / prenatal** (small for gestational age) with recognition in **infancy** (HP:0003593 Infantile onset, 3/3). Pattern is **insidious/chronic**, present from birth.
- **Progression: chronic, lifelong, non-remitting**. Growth failure and developmental delay are evident early; brain atrophy suggests a potentially **progressive/static-with-deterioration** CNS component in at least some individuals. Seizures/epileptic encephalopathy can evolve. No staging system exists.
- **Disease course:** stable-to-slowly-progressive developmental impairment; episodic events = seizures and intercurrent respiratory infections.
- **Remission:** none (no spontaneous or treatment-induced remission; disorder is constitutional).
- **Critical periods:** fetal and early-postnatal neurodevelopment (period of maximal neural-progenitor proliferation) is the key window of vulnerability; interventions are supportive (early developmental/rehabilitation therapy).

9. Inheritance and Population

Epidemiology. Ultra-rare. Prevalence/incidence **not established** — only **3 affected individuals from 2 consanguineous families** reported (34313816).

Likely far fewer than 1 per 1,000,000; no registry data.

Genetic etiology. - **Inheritance: Autosomal recessive** (HP:0000007) — *"ZNF668 deficiency in humans results in a recognizable autosomal recessive disorder"* (34313816).

- **Penetrance:** presumed **complete** for biallelic truncating genotype (all biallelic individuals affected), though n is tiny. - **Expressivity: variable** for secondary features (cardiac, ocular, metabolic, skeletal) across the small cohort; core features (growth failure, microcephaly, DD) are consistent. - **Anticipation:** not applicable (not a repeat-expansion disorder). - **Germline mosaicism:** not reported. - **Founder effect:** none established; variants are family-specific/private. - **Consanguinity: major role** — both families consanguineous; disorder manifests through homozygosity-by-descent. Expected to be enriched in populations with high consanguinity rates. - **Carrier frequency:** truncating alleles are ultra-rare in gnomAD (gene strongly LoF-constrained), so carrier frequency is very low except possibly within specific consanguineous kindreds.

Population demographics. Reported families include Middle Eastern (Saudi; Alkuraya group) and European (French, Angers group) ancestry based on the author cohort (34313816).

Predicted to be found more often in **consanguineous/endogamous populations**. **Sex ratio:** no sex bias expected (autosomal); not quantifiable (n=3). **Age distribution:** affected from birth; survivors are children (natural history into adulthood unknown).

10. Diagnostics

Primary diagnostic modality — genetic testing. - **Recommended approach: trio whole-exome sequencing (WES)** or **whole-genome sequencing (WGS)** in a child with prenatal-onset growth restriction + microcephaly + developmental delay + dysmorphism, interpreted with **homozygosity mapping** / **runs-of-homozygosity** in consanguineous pedigrees (the gene-discovery route used in (P 34313816). WGS adds value for non-coding/structural alleles. **[human clinical]** - **Gene panels:** inclusion in broad **intellectual-disability** / **microcephaly** / **primordial-dwarfism** / **multiple-congenital-anomaly** NGS panels (single-gene *ZNF668* testing otherwise rarely offered). - **Single-gene** / **segregation:** Sanger confirmation and parental carrier testing to confirm biallelic inheritance. - **CMA** / **karyotype** / **FISH:** chromosomal microarray and karyotype are typically **normal** in ZNF668-ZMAND (point/truncating variants); useful mainly to exclude CNVs/aneuploidy. **Caveat:** a 16p11.2 CNV involving *ZNF668* on CMA indicates the separate **16p11.2 microdeletion/microduplication syndrome**, not the recessive single-gene disorder. Mitochondrial DNA and repeat-expansion testing: not indicated.

Supportive clinical tests. - **Neuroimaging (brain MRI):** to document microcephaly, brain malformation, brain atrophy (HP:0012444). Key for phenotyping. - **Functional DDR assay (research/supportive):** patient-fibroblast DNA-damage-repair assays demonstrate the repair defect (P 34313816) — a functional biomarker of pathogenicity, not a routine clinical test. **[human clinical** / **in vitro]** - **EEG:** for seizures/epileptic encephalopathy. - **Echocardiography:** for atrial septal defect. - **Ophthalmologic exam:** for chorioretinal coloboma/deep-set eyes. - **Growth monitoring, bone densitometry** (osteopenia), **glucose/HbA1c** (diabetes surveillance), **feeding/respiratory assessment**. - **Biomarkers:** no specific circulating biochemical biomarker; diagnosis is molecular. LOINC-coded genetic report; no disease-specific metabolite.

Clinical criteria / **differential diagnosis.** No formal diagnostic criteria (ultra-rare). Diagnosis = compatible phenotype + biallelic LoF *ZNF668*. **Differential diagnosis:** other autosomal-recessive **microcephalic primordial dwarfism** / **microcephaly-DDR syndromes** and genome-instability disorders — e.g., **Seckel syndrome, microcephalic osteodysplastic primordial dwarfism type II (MOPD II)**, MCPH spectrum, Cornelia de Lange syndrome, and other DDR disorders

(P 19473885 reviews this class). Distinguishing feature: biallelic *ZNF668* truncation with an in-vitro DDR defect and the specific facial gestalt (large ears + prominent nose + deep-set eyes).

Screening. Not part of newborn screening. **Cascade carrier testing** of relatives and **prenatal/preimplantation testing** available once familial variants are known.

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11. Outcome / Prognosis

- **Survival / mortality:** not quantified. No survival statistics (n=3; natural history into adulthood unknown). Potential early-life morbidity/mortality risk from severe neurodevelopmental impairment, seizures, feeding difficulty, and recurrent respiratory infections (inferred by analogy to severe microcephaly/DDR syndromes).
- **Morbidity / disability: severe, lifelong**. Severe global developmental delay and intellectual disability (ICF: profound limitations in learning, communication, mobility, self-care), hypotonia ± spasticity, epilepsy. High dependency for daily activities.
- **Quality of life:** profoundly impacted; no formal EQ-5D/SF-36/PROMIS data.
- **Complications:** epileptic encephalopathy, aspiration/feeding difficulties, recurrent lower respiratory infections, cardiac (ASD), osteopenia/fracture risk, diabetes mellitus (variable), visual impairment (coloboma).
- **Recovery potential:** none for the core neurodevelopmental deficit (constitutional); supportive care improves function/comfort.
- **Prognostic factors / biomarkers:** severity of microcephaly/brain malformation and epileptic encephalopathy likely predict worse neurodevelopmental outcome (inferred). No validated prognostic molecular biomarker. **Theoretical long-term consideration:** because *ZNF668* is a tumor suppressor somatically, whether constitutional biallelic loss alters lifetime cancer risk is **unknown and unstudied** — flagged as a research gap.

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12. Treatment

No disease-specific or curative therapy exists. Management is **symptomatic and supportive/multidisciplinary** (NCIT:C15277 Supportive Care; NCIT:C15688 Symptomatic Treatment).

- **Pharmacotherapy:** none targeting ZNF668. **Antiseizure medications** for epilepsy (NCIT:C264 Anticonvulsant Agent). Standard management of diabetes mellitus if present (NCIT:C159044 Antidiabetic Agent). No pharmacogenomic guidance specific to this disorder.
- **Advanced therapeutics (gene/cell/RNA/targeted/immuno): none**; no gene therapy, ASO, or targeted therapy in development. Not applicable.
- **Surgical / interventional:** as indicated for complications — e.g., **gastrostomy** for feeding failure (NCIT:C157826), **ASD repair** if hemodynamically significant (NCIT:C51818 Cardiac Surgery), strabismus/ophthalmic or orthopedic interventions as needed.
- **Supportive & rehabilitative:** nutritional support for failure to thrive; **physical, occupational, and speech/language therapy** (NCIT:C15607 Physical Therapy; NCIT:C15630 Occupational Therapy; NCIT:C15312 Speech Therapy); respiratory care/infection prophylaxis; bone-health management for osteopenia; developmental/special-education support.
- **Experimental treatments / trials: none registered** (no ClinicalTrials.gov NCT entries for ZNF668-related disorder).
- **Treatment outcomes / adverse events:** no disease-specific response data.
- **Treatment strategy:** individualized, guideline-based management of each manifestation; genotype-guided care limited to confirming diagnosis and enabling family counseling.

13. Prevention

- **Primary prevention:** not possible for the constitutional disease. At the family/population level, **genetic counseling about consanguinity** and recurrence risk is the principal preventive measure.
- **Secondary prevention (early detection):** early genetic diagnosis enables early developmental intervention and surveillance (seizures, cardiac, ophthalmologic, metabolic/diabetes, bone health).

- **Tertiary prevention:** proactive management to prevent complications — seizure control, aspiration/respiratory-infection prevention, nutritional optimization, bone-health measures.
- **Immunization:** routine childhood immunizations recommended (reduce respiratory-infection morbidity); no disease-specific vaccine.
- **Genetic screening / reproductive options: carrier (cascade) testing** of relatives; **prenatal diagnosis** and **preimplantation genetic testing (PGT-M)** for at-risk couples once familial biallelic variants are identified (ACMG/ACOG frameworks).
- **Counseling:** autosomal-recessive **25% recurrence risk** for carrier couples; emphasize consanguinity risk.
- **Public-health / environmental / prophylaxis:** not applicable beyond general measures.

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14. Other Species / Natural Disease

- **Taxonomy / natural disease:** No naturally occurring animal disease described (OMIA: none). Human disease only (**NCBITaxon:9606**).
- **Orthologous genes (NCBI Gene / ortholog symbols):** mouse **Zfp668** (Mus musculus, NCBITaxon:10090); conserved across mammals and vertebrates (the C2H2 zinc-finger architecture is evolutionarily conserved). Rat *Zfp668* (NCBITaxon:10116) predicted.
- **Veterinary relevance:** none known.
- **Comparative biology:** the **DNA-damage-response/HR-repair and p53–MDM2 pathways** in which ZNF668 acts are deeply conserved; disease-mechanism conservation is expected, but the specific syndrome has not been reported in other species. **Evolutionary conservation** of the DDR machinery underpins the expectation that loss would be deleterious across species.
- **Zoonotic / cross-species transmission:** not applicable (genetic disease).

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15. Model Organisms

- **Available models: No published vertebrate disease model** (no reported *Zfp668* knockout mouse phenotype). **IMPC** has **no significant phenotype annotations** for mouse *Zfp668* at

time of writing (not yet phenotyped or no reported significant hit) — i.e., no mouse model demonstrating phenotype recapitulation. **[computational/database]**

- **In-vitro / cellular models (the actual evidence base):**
- **Patient-derived skin fibroblasts** — demonstrate the DNA-damage-repair defect (P 34313816). **[human clinical cells]**
- **U2OS osteosarcoma cells with ZNF668 knockdown** — reduced Tip60–H2AX interaction, impaired H2AX acetylation/chromatin relaxation, defective HR, reduced survival after IR/UV (P 23777805). **[in vitro]**
- **Human mammary epithelial cells** — ZNF668 knockdown sufficient to transform cells; ZNF668 suppresses proliferation/tumorigenicity; impaired DNA-damage-induced p53 stabilization (P 21852383). **[in vitro]**
- **Model types needed/applicable:** mouse knockout/conditional (brain-/growth-specific), zebrafish *znf668* morphant/CRISPR, and **patient iPSC-derived neural organoids** would be informative for recapitulating microcephaly/neurodevelopment — **none yet reported** (research gap).
- **Phenotype recapitulation / limitations:** current cellular models recapitulate the **molecular DDR defect** (repair, p53, chromatin) but **not** the organismal phenotype (microcephaly, growth failure, dysmorphism). No model captures the full clinical syndrome.
- **Resources:** MGI (mouse *Zfp668*), IMPC, Alliance of Genome Resources, Cellosaurus (U2OS); IMSR/KOMP for potential alleles.

Supported vs. Refuted Hypotheses

Supported. - NEDGDF (MONDO:0859350/OMIM 620194) is caused by **biallelic truncating LoF variants in *ZNF668***, autosomal recessive (34313816). ✔ - Mechanism is a **DNA-damage-repair defect** via impaired Tip60/H2AX-acetylation-dependent HR repair and p53–MDM2 dysregulation (PMIDs 23777805, 21852383, 34313816). ✔ - *ZNF668* is strongly **LoF-constrained** (gnomAD pLI 0.999), consistent with an essential dosage-sensitive gene. ✔ - Core invariant phenotype = prenatal-onset growth restriction + microcephaly + severe DD + facial dysmorphism (HPO frequencies, 34313816). ✔

Refuted / not supported. - Not an environmental, infectious, or acquired disorder. ✘ - Not a dominant/de-novo disorder (recessive; heterozygous carriers unaffected). ✘ - No evidence for a repeat-expansion/anticipation mechanism. ✘

Limitations and Future Directions

Limitations. Evidence rests on a **single defining publication with only 3 patients**; all phenotype frequencies have denominators of 2–3 and are provisional. No natural-history, survival, QoL, or treatment-outcome data. No animal model and no patient omics datasets. Exact variant-level nomenclature (cDNA/protein HGVS) was not retrievable from abstract-level sources in this analysis and should be confirmed from the primary paper/ClinVar. Orphanet/ICD-specific codes not yet assigned. Potential constitutional cancer-risk implications of biallelic tumor-suppressor loss are unknown.

Future directions. (1) Identify additional families via GeneMatcher to refine the phenotype spectrum and penetrance; (2) deposit/curate variants in ClinVar; (3) generate *Zfp668* mouse/zebrafish and patient iPSC neural-organoid models to test microcephaly recapitulation; (4) characterize patient-cell transcriptomic/epigenomic signatures and p53-pathway status; (5) assess lifetime tumor risk given ZNF668's tumor-suppressor role; (6) establish surveillance guidelines (seizure, cardiac, ophthalmologic, metabolic, bone).

Key References (PMIDs)

- **34313816** — Alsaif et al., 2021. *ZNF668 deficiency causes a recognizable disorder of DNA damage repair.* (Disease-defining; 3 patients; AR; DDR defect in fibroblasts; proposes ZNF668-related ZMAND.) **[human clinical]**
- **21852383** — Hu et al., 2011. *ZNF668 functions as a tumor suppressor by regulating p53 stability and function in breast cancer.* (Nucleolar protein; p53/MDM2 regulation.) **[in vitro]**
- **23777805** — Hu et al., 2013. *Zinc finger protein 668 interacts with Tip60 to promote H2AX acetylation after DNA damage.* (Chromatin relaxation; HR repair.) **[in vitro]**
- **37240013** — Okuno et al., 2023. *Reduced expression of ZNF668 in bladder cancer and its invasiveness.* **[in vitro/tumor]**
- **19473885** — Kerzendorfer & O'Driscoll, 2009. *Human DNA damage response and repair deficiency syndromes.* (Context: microcephalic DDR-disorder class.) **[review]**
